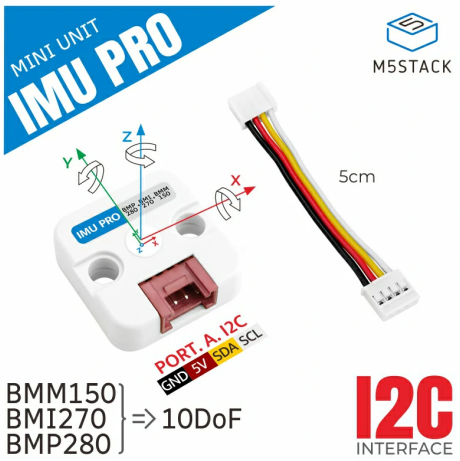
Unit Mini IMU-Pro

SKU:U171













Description

Unit Mini IMU-Pro is a versatile 10-degree-of-freedom inertial motion unit that integrates a six-axis attitude sensor (BMI270), a three-axis geomagnetic sensor (BMM150), and an atmospheric pressure sensor (BMP280). It measures and detects the acceleration and angular velocity of objects, the direction and intensity of the geomagnetic field, and atmospheric pressure. This unit communicates via the I2C interface and features an onboard toggle switch to change the I2C address of the BMI270. It is used in navigation, automation control, robotics, aircraft navigation, weather detection, and geolocation.

Features

- Multi-functional sensor unit
- High-precision measurement
- I2C communication interface
- Compact design
- Compatible with programming platforms: UIFlow, Arduino, etc.

Includes

- 1 x Unit Mini IMU-Pro
- 1 x HY2.0-4P Grove cable (5cm)

Applications

- Navigation
- Automation control
- Robotics
- Aircraft navigation
- Geolocation

Specifications

Specification	Parameter
6-Axis Attitude Sensor (BMI270)	Accuracy: 0.05% (acceleration), 0.05°/s (angular velocity) I2C address @0x68 (default)
3-Axis Geomagnetic Sensor (BMM150)	Accuracy: 0.3 μT I2C address @0x10
Barometric Pressure Sensor (BMP280)	Accuracy: ±1°C (temperature), ±1 Pa (pressure) I2C address @0x76
Product Size	24.0 x 24.0 x 8.0mm
Product Weight	3.1g
Package Size	138.0 x 93.0 x 9.0mm
Gross Weight	6.6g

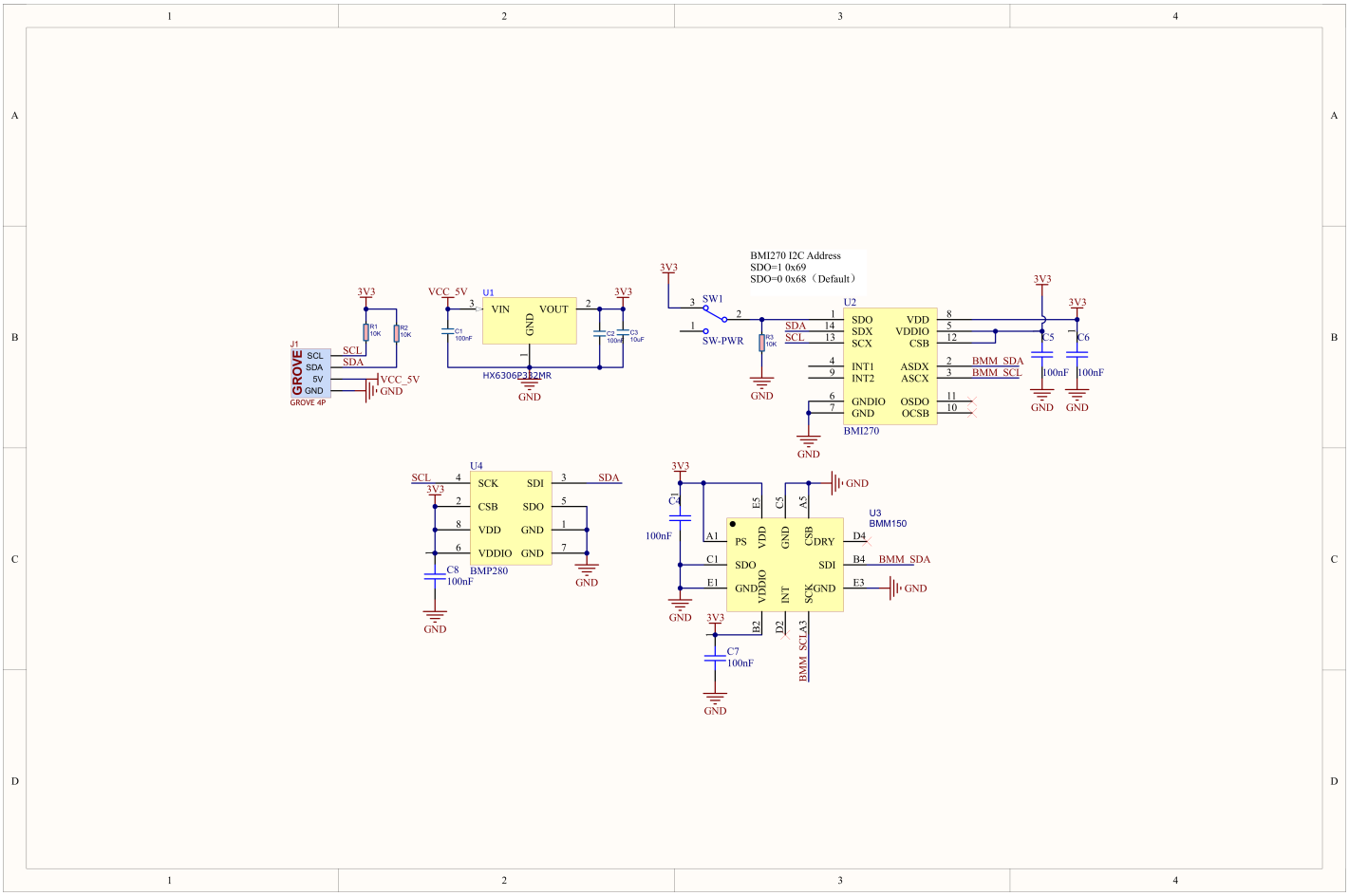
Learn

BMM150 Magnetic Field Interference

Products with magnets may interfere with the BMM150 magnetic field sensor, causing abnormal readings. When used with an M5 master control device containing a magnet, the magnet needs to be removed, and at the same time, the BMM150 sensor should be kept away from strong magnetic fields.

Schematics

- Unit Mini IMU-Pro Schematics PDF

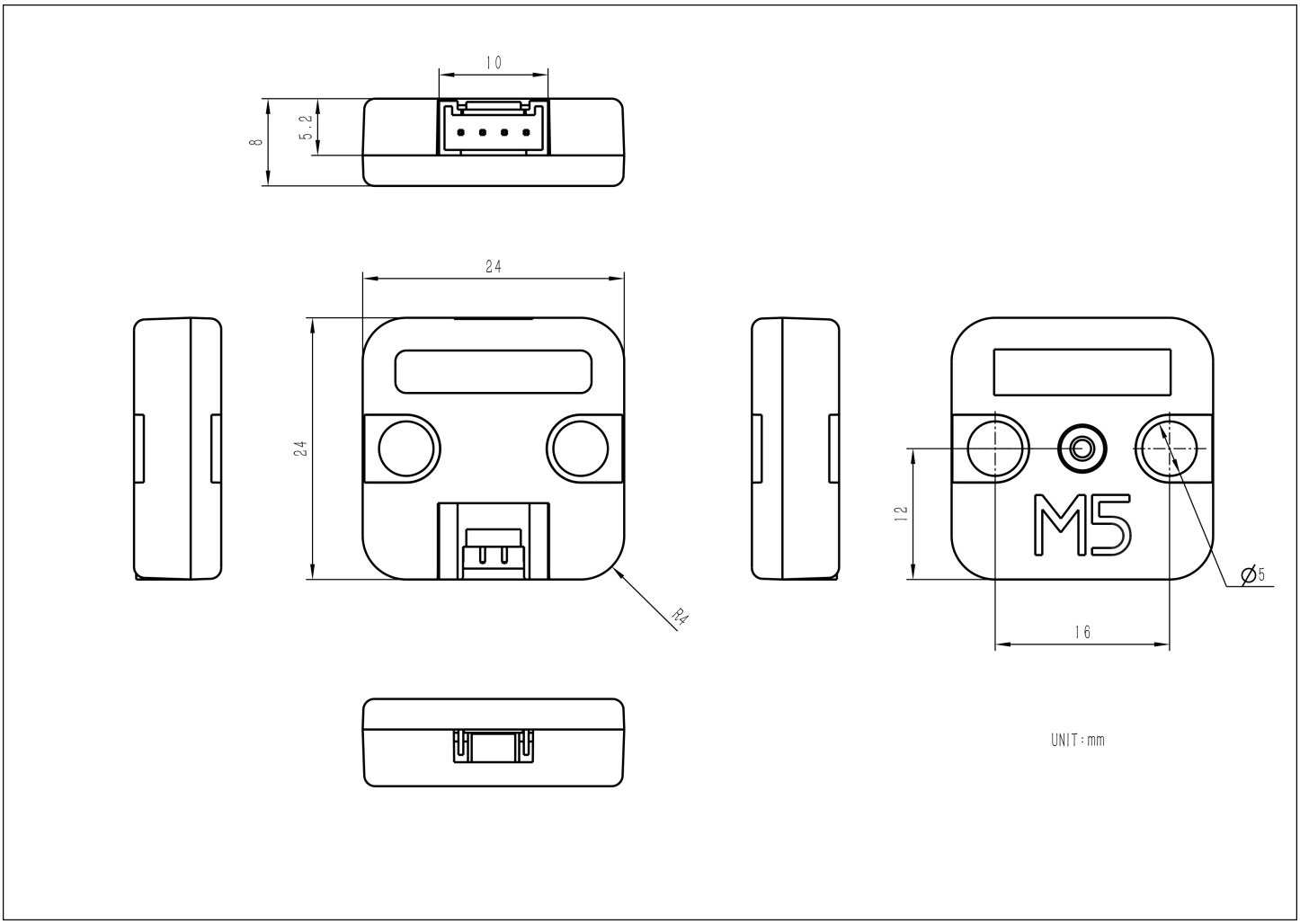


PinMap

Unit Mini IMU-Pro

HY2.0-4P	Black	Red	Yellow	White
PORT.A	GND	5V	SDA	SCL

Model Size



Datasheets

- BMI270
- BMM150
- BMP280

Softwares

Arduino

- Unit Mini IMU-Pro Arduino Library
- Unit Mini IMU-Pro Get Raw Data
- Unit Mini IMU-Pro Get XYZ Value AHRS Example

UiFlow1

- Unit Mini IMU-Pro UiFlow1 Docs

UiFlow2

- Unit Mini IMU-Pro UiFlow2 Docs

Video

- Unit Mini IMU-Pro Application Example

U171 IMU Pro 视频.mp4